

CY 550: Bioinspired Materials

Course Code	CY 550
Course Name	Bioinspired Materials
L-T-P-C	3-0-0-3
Prerequisites	IC 130 and IC 241 /or permission from instructor
Intended for	B.Tech. (all branches), M.Sc., M.Tech., M.S. and Ph.D.
Distribution	Elective
One-time approval	recommended by the 9th BoA (Oct 2015)

Course Contents

- **Fundamental Principles**

Biomimicry introduction; design concepts; typical examples; intelligent life; multiscale hierarchical structures; heterogeneous interfaces; high strength materials; biomimetic strategies and materials processing [8 Lectures]

- **Concepts and Processes**

Bioinspired synthesis of nanostructures; supramolecular chemistry; self-assembly; controlled assembly; biomineralization; stimuli-responsive behavior; self-replication; self-healing; self-cleaning; anti-reflection; anti-fouling; wettability; adhesion; scaffolds; multi-functionality; sensing; transduction; camouflage; swarm intelligence; fabrication and replication of biological systems [12 Lectures]

- **Technological Applications**

Artificial photosynthesis; renewable energy; biofuels; photovoltaics; catalysis; ion channels; medical diagnostics; drug delivery; tissue engineering; prosthetics; electronics; photonics; optics; fog harvesting; water filtration; defense; robotics; aeronautics; other new technological applications; review of current research. [12 Lectures]

- **Investigation Tools** [10 Lectures]

High resolution optical microscopy; scanning electron microscopy; transmission electron microscopy; scanning probe microscopies; x-ray absorption spectroscopy; x-ray tomography; x-ray scattering; x-ray photoelectron spectroscopy; time-of flight secondary ion mass spectrometry; neutron scattering; single molecule probes; theoretical modeling and computer simulations

Text Books

1. L. Jiang, L. Feng, **Bioinspired Intelligent Nanostructured Interfacial Materials**, World Scientific Publishing Company, 2010.
2. S. Mann (Ed.), **Biomimetic Materials Chemistry**, Wiley-VCH, 1995.

Readings

1. Y. Zhou, **Bio-Inspired Nanomaterials and Nanotechnology**, Nova Science Publishers, 2009.
2. C. S. S. R. Kumar (Ed.), **Biomimetic and Bioinspired Nanomaterials**, Wiley-VCH, 2010.
3. Y. Bar-Cohen, **Biomimetics: Biologically Inspired Technologies**, CRC Press, 2005.
4. Youtube Channel: <http://www.youtube.com/functionalsurfaces>